Advanced Algebra

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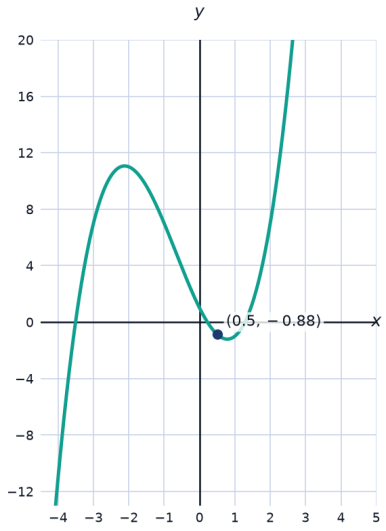
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying rational expressions

Combine into a single fraction: $\frac{2}{x+1} + \frac{3}{x-1}$.

2 [Foundational] • Target: Polynomial division, remainder & factor theorems

Use the remainder theorem to find the remainder when $f(x) = x^3 + 2x^2 - 5x + 1$ is divided by $(x - 2)$.



3 [Foundational] • Target: Polynomial division, remainder & factor theorems

Given $f(x) = x^3 + 3x^2 - 4x - 12$:

a Use the factor theorem to show that $(x + 3)$ is a factor of $f(x)$.

b Hence factor $f(x)$ completely.

4 [Foundational] • Target: Polynomial division, remainder & factor theorems

Divide $x^3 - 2x^2 - 5x + 6$ by $(x - 3)$ using polynomial long division. State the quotient and remainder.

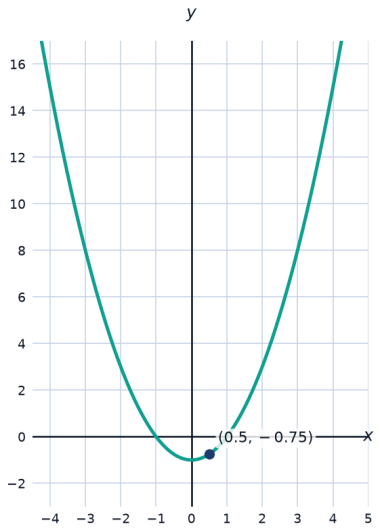
5 [Foundational] • Target: Exponential equations

Solve $2^x = 15$ for x, giving your answer to three decimal places.

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6 [Foundational] • Target: Nonlinear systems of equations

Solve the system $y = x^2 - 1$ and $y = 2x + 2$.



7 [Foundational] • Target: Systems of inequalities and linear programming

A feasible region is bounded by the constraints $x \geq 0$, $y \geq 0$, $x + y \leq 6$, and $2x + y \leq 10$. Find the vertices of the feasible region.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Systems of inequalities and linear programming

Using the feasible region from the previous question, find the maximum value of $P = 3x + 2y$ over the region, and state where it occurs.

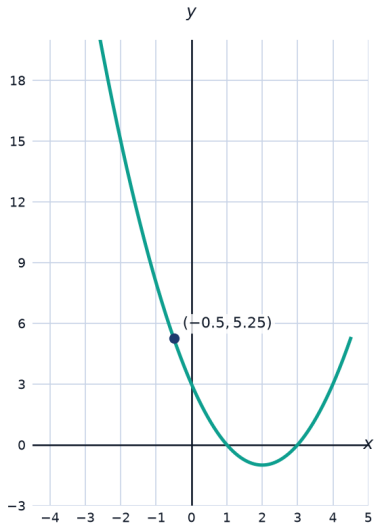
9 [Intermediate] • Target: A well-designed word problem: production planning at a furniture workshop

This question has three parts. A furniture workshop makes tables and chairs. Each table requires 4 hours of carpentry and 2 hours of finishing; each chair requires 1 hour of carpentry and 1 hour of finishing. Each week the shop has at most 32 hours of carpentry time and 20 hours of finishing time available. Each table earns a profit of 70 dollars and each chair earns a profit of 30 dollars. Let x be the number of tables and y the number of chairs produced in a week. (a) Write the system of inequalities describing the constraints (including $x \geq 0$ and $y \geq 0$). (b) Find the vertices of the feasible region. (c) Determine the number of tables and chairs that maximizes the weekly profit $P = 70x + 30y$, and find the maximum profit.

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10 **[Intermediate]** • **Target: Simplifying rational expressions**

Solve the nonlinear system of equations: $\{y = x^2 - 4x + 3; 2x + y = 6$ Find all points of intersection (x, y).



11 **[Intermediate]** • **Target: Absolute value equations**

Find all real solutions to the rational equation $\frac{2}{x-3} - \frac{3}{x+1} = \frac{8}{x^2-2x-3}$.

12 **[Intermediate]** • **Target: Polynomial division, remainder & factor theorems**

When the polynomial $P(x) = 2x^3 - kx^2 + 5x - 7$ is divided by $(x - 2)$, the remainder is 11. Find the value of k.

13 **[Intermediate]** • **Target: Logarithmic equations**

Solve the logarithmic equation $\log_2(x + 4) - \log_2(x) = 3$ for x.

14 **[Intermediate]** • **Target: Exponential equations**

When the polynomial $P(x) = 2x^3 - kx^2 + 5x - 6$ is divided by $x - 2$, the remainder is 8. What is the value of the constant k?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

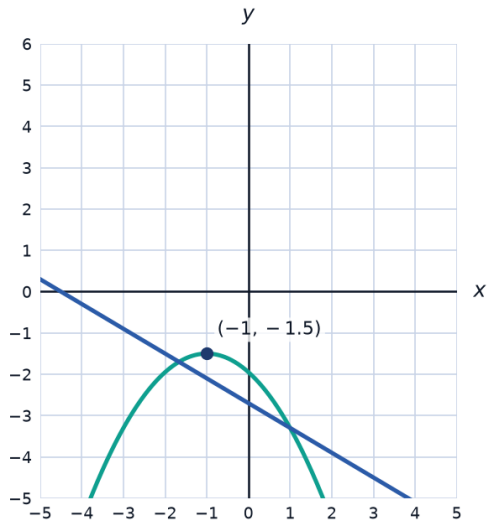
15 **[Challenging]** • **Target: Nonlinear systems of equations** • **Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic**

Solve the exponential equation $3^{2x-1} = 5^{x+2}$ for x in terms of natural logarithms.

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16 [Challenging] • Target: Rational equations and extraneous solutions • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

On the EST II Math exam, Advanced Algebra and Functions together account for 40% of the questions. In a 50-question test, if a student targets an accuracy rate of 90% in these two high-yield domains, how many questions from these domains can the student afford to miss at most?



17 [Challenging] • Target: Systems of inequalities and linear programming • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve the rational equation for x: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

18 [Challenging] • Target: A well-designed word problem: production planning at a furniture workshop • Cross-Lesson: 5.01 Trigonometry

Solve the logarithmic equation $\log_3(x+6) + \log_3(x) = 3$ and eliminate any extraneous solutions.

19 [Challenging] • Target: Simplifying rational expressions • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Find all solutions of $\cos(2\theta) = \cos\theta$ in the interval $0^\circ \le \theta \le 360^\circ$.

20 [Challenging] • Target: Absolute value equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

Let $f(x) = \frac{2x+1}{x-3}$ for $xeq3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Simplifying rational expressions

Solve the system $y = x^2 + 4x + 1$ and $y = -5x - 7$. Find the x-coordinates of the intersection points.

a. $-8, -3$

b. $-8, -1$

c. $-6, -1$

d. $8, 1$

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22 [Foundational] • Target: Absolute value equations

Using the Factor Theorem, find the remainder when $f(x) = x^3 - x^2 - 6x - 4$ is divided by $(x + 2)$.

a. -4

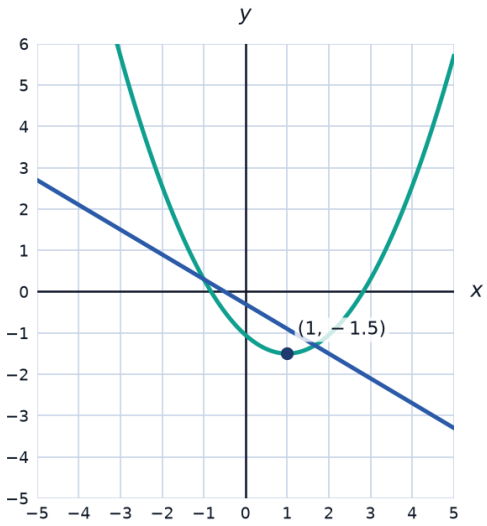
b. -9

c. 1

d. 5

23 [Foundational] • Target: Polynomial division, remainder & factor theorems

In the system of linear equations below, k is a real constant. $(k - 1)x + 6y, = 18; 2x + ky, = 12$ For which value of k does the system have infinitely many solutions (x, y)?



a. -4

b. -3

c. 3

d. 4

24 [Foundational] • Target: Logarithmic equations

When $f(x) = x^2 + 3x + 5$ is divided by $(x - 3)$, what is the remainder?

a. 28

b. 26

c. 19

d. 23

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25 [Foundational] • Target: Exponential equations

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. -9

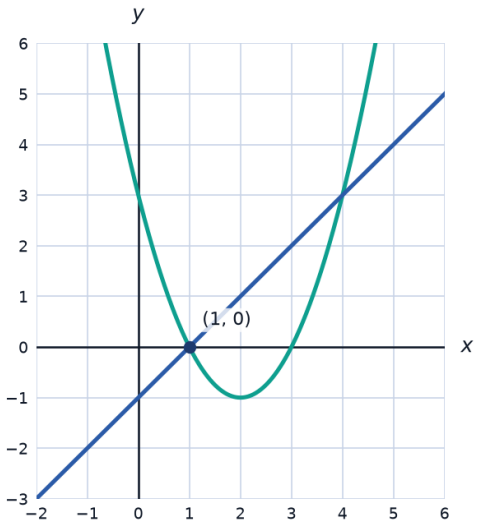
b. -4

c. 4

d. 9

26 [Foundational] • Target: Nonlinear systems of equations

A quadratic function is given by $f(x) = x^2 - 4x + 3$. The graph of this function is a parabola. Which of the following points lies on the graph of $f(x)$?



a. $(1, 0)$

b. $(0, 1)$

c. $(2, 3)$

d. $(4, 4)$

27 [Foundational] • Target: Rational equations and extraneous solutions

In the standard (x, y) coordinate plane, the system of equations $\{y = |x^2 - 4|; y = 2x + k$ has exactly 3 distinct real solutions (x, y). What is the sum of all possible values of the real constant k?

a. 4

b. 5

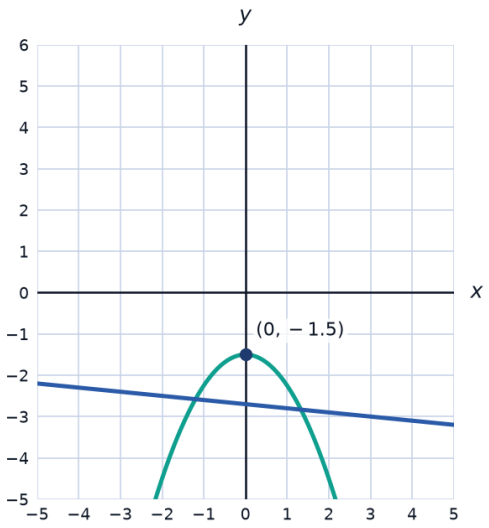
c. 9

d. 11

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28 [Foundational] • Target: Systems of inequalities and linear programming

Consider the function $h(x) = ax^2 + bx + c$. Given that $h(1) = 6$, $h(-1) = 10$, and the graph of $y = h(x)$ has a vertex at $(3, -10)$, find the value of $a + b + c$.



a. 12

b. 6

c. 10

d. 2

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A well-designed word problem: production planning at a furniture workshop

Consider the function $g(x) = ax^2 + bx + c$. The graph of $y = g(x)$ is a parabola that passes through the points $(1, 5)$, $(2, 12)$, and $(3, 21)$. If the vertex of the parabola is at (h, k), what is the value of h?

a. $\frac{5}{2}$

b. 2

c. $-\frac{5}{2}$

d. -2

30 [Intermediate] • Target: Simplifying rational expressions

Consider the function $f(x) = ax^2 + bx + c$. The graph of $y = f(x)$ passes through the points $(1, 2)$, $(2, 7)$, and $(3, 14)$. Find the value of a.

a. 1

b. 2

c. 3

d. 4

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31 [Intermediate] • Target: Absolute value equations

Consider the function $g(x) = x^2 - ax + b$. If the graph of $y = g(x)$ has a minimum value of -1 at $x = 3$, find the values of a and b.

a. $a = 6, b = 8$

b. $a = -6, b = 8$

c. $a = 6, b = -1$

d. $a = -6, b = -1$

32 [Intermediate] • Target: Polynomial division, remainder & factor theorems

If $f(x) = x^2 + 1$ and $g(x) = -x - 1$, find the sum of all solutions to $(f \circ g)(x) = 5$.

a. -2

b. 4

c. -6

d. -10

33 [Intermediate] • Target: Logarithmic equations

When $f(x) = x^4 + x^3 + 5x^2 - 5x - 1$ is divided by $(x + 3)$, the quotient is a cubic $q(x)$. What is the sum of the coefficients of $q(x)$?



a. -37

b. -33

c. -23

d. -28

34 [Intermediate] • Target: Exponential equations

If y varies directly as x and $y = 12$ when $x = 3$, find y when $x = 7$.

a. 21

b. 24

c. 28

d. 25

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35 [Intermediate] • Target: Nonlinear systems of equations

For $x > 0$, solve $x + \frac{10}{x} = 7$ and give the larger root.

a. 9

b. 5

c. -2

d. 11

36 [Intermediate] • Target: Rational equations and extraneous solutions

Consider the function $g(x) = 2x + 3$. What is the value of $g(4)$?

a. 7

b. 8

c. 11

d. 14

37 [Intermediate] • Target: Systems of inequalities and linear programming

Divide $f(x) = x^3 + 3x^2 + 2x - 8$ by $(x - 3)$. What is the constant term of the quotient?

a. 18

b. 27

c. 25

d. 20

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: production planning at a furniture workshop • Cross-Lesson: 5.01 Trigonometry

Two similar solids have a linear scale factor of $2:1$. If the surface area of the smaller solid is 12, find the surface area of the larger solid.

a. 14

b. 2

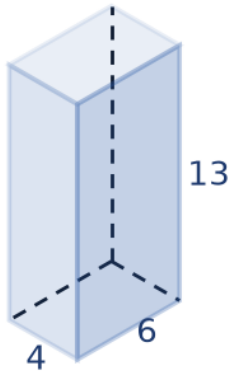
c. 13

d. 3

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39 [Challenging] • Target: Simplifying rational expressions • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Two similar solids have a linear scale factor of $1:4$. If the volume of the smaller solid is 13, find the volume of the larger solid.



a. 443

b. 655

c. 832

d. 193

40 [Challenging] • Target: Absolute value equations • Cross-Lesson: 7.01 Solid Geometry & Similarity

What is the larger solution to $|x - 5| = 3$?

a. 10

b. 8

c. 2

d. 14

41 [Challenging] • Target: Polynomial division, remainder & factor theorems • Cross-Lesson: 8.01 Statistics & Probability

The line $y = -12$ intersects the circle $x^2 + y^2 = 169$ at two points. What is the distance between the two intersection points?

a. 10

b. 13

c. 16

d. 5

42 [Challenging] • Target: Logarithmic equations • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

Simplify $\frac{x^2 - 16}{x - 4}$, then evaluate the result at $x = 8$.

a. 6

b. 14

c. 11

d. 12

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43 [Challenging] • Target: Exponential equations • Cross-Lesson: 3.01 Sequences, Series & Complex Numbers

The line $y = 4$ meets the circle $x^2 + y^2 = 25$ at a point with positive x-coordinate. Find that x-coordinate.

a. -2

b. 7

c. 11

d. 3

44 [Challenging] • Target: Nonlinear systems of equations • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Two similar solids have a volume ratio of $125 : 1$. Find the linear (side-length) scale factor.

a. 1:9

b. 1:10

c. 1:5

d. 1:7

45 [Challenging] • Target: Rational equations and extraneous solutions • Cross-Lesson: 5.01 Trigonometry

The line $y = -2x + 2$ intersects the parabola $y = x^2 + x + 4$ at two points. What is the sum of the x-coordinates of the intersection points?

a. -3

b. 1

c. -12

d. 3

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